

CPSC 2030 Assignment 5: jQuery DOM Traversal and Manipulation

- I have provided HTML and CSS code for this assignment. Please do not modify this code unless specifically directed to in the instructions!

For Tasks 1 and 2, Write your jQuery code in a file called **myscript.js**.

Task 1: Add to Gallery Buttons



- Write a jQuery event listener that listens for a click on the “Add to top” button and adds the image generated to the top of the section with class “image-gallery”.
- Write a jQuery event listener that listens for a click on the “Add to bottom” button and adds the image generated to the bottom of the section with class “image-gallery”.

Hint: You can use any of the following jQuery methods to add elements before or after other elements:

- Appending: [append\(\)](#) , [appendTo\(\)](#)
- Prepending: [prepend\(\)](#), [prependTo\(\)](#)
- Adding elements [after\(\)](#) and [before\(\)](#) an element

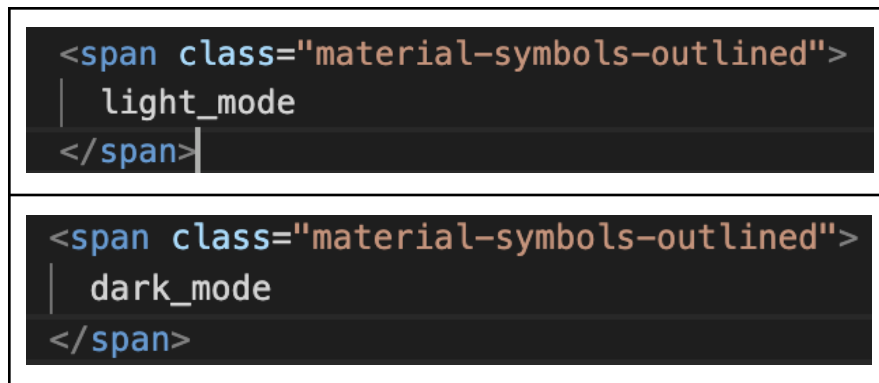
Note that there is an image container near the top of the page and another one near the bottom. Your jQuery code should be **generalized to work with any widget on the page**. This means that you should use the **this** keyword to make sure you're working with the element that triggered the event (rather than writing another set of event listeners for the bottom image container).

Task 2: Theme Toggle



Write a jQuery event listener that listens for a click on the “Toggle Theme” button and toggles between two states - dark and light themes.

- a. I've already added a CSS class called "darkMode" that you can toggle between.
 - i. Be default, light mode is selected. When the user first clicks the toggle, you can add the "darkMode" class on the <body> tag to enable dark mode.
 - ii. The second time the toggle theme button is clicked, you can remove the "darkMode" class to go back to light mode.
 - iii. The user should be able to switch between these modes indefinitely.
- b. To toggle between the Sun and Moon icons in the "Toggle Theme" button, you need to change the text inside the with class="material-symbols-outlined"



- i. Hint: You can use the [.text\(\)](#) jQuery method to change the text inside an HTML element.
- c. Finally, you need to animate the theme change (as shown in the demo).
 - i. Hint: One approach is to use the transition property for the CSS properties that are changing when the darkMode class is toggled (background-color and color) and use an appropriate transition-duration.

Task 3: Bonus Challenge Question (optional exercise):

As an optional bonus task, repeat the assignment using only vanilla JavaScript (no jQuery)

For Task 3, Save your jQuery code in a file called **myscript2.js**.

Hand in

- Rename your folder to **a5_firstname_lastname**, zip it, and hand it into the Assignment 5 folder.

Checklist

- [5 marks] Task 1: Add to Gallery Buttons
- [5 marks] Task 2: Theme Toggle
- [1 bonus point] Task 3: Reimplement Tasks 1 and 2 using only JavaScript

Total: 10

Acknowledgements and References:

- jQuery API <http://api.jquery.com>